Splunk for SOC Analysts

Definitions, SPL Commands, Security Detections & Investigation Workflows

1. What is Splunk?

Splunk is a SIEM and data analytics platform used to collect, search, monitor, correlate, and analyze machine-generated data such as Windows logs, Sysmon events, firewall logs, network logs, and application logs.

In a SOC, Splunk helps analysts monitor security events, investigate suspicious activity, detect attacks, create alerts, build dashboards, perform threat hunting, and investigate incidents.

2. Important Splunk Components

Search Head: Interface where analysts search and analyze data using SPL.
Indexer: Stores and processes incoming data.
Universal Forwarder: Lightweight agent used to forward endpoint logs to Splunk.
Heavy Forwarder: Full Splunk instance capable of additional processing before forwarding.
Index: Logical storage location for indexed data.
Sourcetype: Identifies the format/type of incoming data.
Source: Identifies where the data originated.
Host: Identifies the system that generated the event.

3. Basic SPL

SPL stands for Search Processing Language. It is used to search, filter, transform, and analyze Splunk data.

Basic Search

```
index=*

index=windows

index=windows EventCode=4625
```

4. Search Commands

search: Filters events.
fields: Selects fields.
table: Displays selected fields.
sort: Sorts results.
dedup: Removes duplicate values.

Search Command Examples

```
index=windows EventCode=4625
| fields _time, User, src_ip, EventCode


index=windows EventCode=4625
| table _time User src_ip EventCode


index=windows EventCode=4625
| sort - _time


index=windows
| dedup User
```

5. Filtering Commands

where: Filters results using conditions.
eval: Creates or modifies fields.
rename: Changes field names.
rex: Extracts information using regular expressions.

Filtering Examples

```
index=windows
| where EventCode=4625 AND User!="Administrator"
```

```
index=windows
| eval Status=if(EventCode=4625,"Failed Login","Other")
| table _time User EventCode Status
```

```
index=windows
| rename src_ip AS "Source IP"
| table _time User "Source IP"
```

```
index=windows
| rex "User:\s+(?<Username>\w+)"
```

6. Statistical Commands

stats: Performs statistical calculations.
top: Displays the most common values.
chart: Creates statistical tables.
timechart: Analyzes events over time.

Statistical Examples

```
index=windows EventCode=4625
| stats count
```

```
index=windows EventCode=4625
| stats count BY User
```

```
index=windows EventCode=4625
| stats count BY src_ip
```

```
index=windows EventCode=4625
| top User
```

```
index=windows
| chart count BY EventCode
```

```
index=windows
| timechart count BY EventCode
```

7. Time-Based Searching

index=windows earliest=-24h
index=windows earliest=-24h latest=now
index=windows earliest=-7d

Time ranges allow analysts to investigate activity within a defined period and build event timelines.

8. Windows Security Event 4624 – Successful Logon

Event ID 4624 indicates a successful authentication.

Detection Query

```
index=windows EventCode=4624
| table _time User Logon_Type src_ip
```

9. Windows Security Event 4625 – Failed Logon

Event ID 4625 indicates a failed authentication attempt. It is commonly used for brute-force and password-spraying investigations.

Detection Query

```
index=windows EventCode=4625
| table _time User src_ip EventCode
```

10. Brute-Force Detection

A simple detection can identify repeated failed authentication attempts from a source IP or against a user account.

Detection Query

```
index=windows EventCode=4625
| stats count AS Failed_Attempts BY src_ip User
| where Failed_Attempts >= 5
| sort - Failed_Attempts
```

Investigation Steps

1. Identify the source IP.
2. Identify targeted accounts.
3. Review the time range.
4. Check whether a successful login occurred afterward.
5. Investigate the source IP with threat intelligence.
6. Determine whether the activity is legitimate or malicious.

11. Successful Login After Multiple Failures

This pattern may indicate a successful authentication after repeated failed attempts.

Detection Query

```
index=windows (EventCode=4625 OR EventCode=4624)

| stats count(eval(EventCode=4625)) AS Failed, count(eval(EventCode=4624)) AS Successful BY User src_ip

| where Failed >= 5 AND Successful >= 1
```

12. RDP Login Detection

Windows RDP commonly uses Logon Type 10. Analysts can investigate unusual source IPs, users, login times, and unexpected remote access.

Detection Query

```
index=windows EventCode=4624 Logon_Type=10
| table _time User src_ip Workstation_Name Logon_Type
```

13. PowerShell Detection

PowerShell execution can be monitored through Windows or Sysmon logs. Analysts should investigate encoded commands, downloads, hidden execution, and unusual parent processes.

Detection Query

```
index=sysmon EventCode=1
| search Image="*\\powershell.exe"
| table _time User Image CommandLine ParentImage
```

14. Suspicious Process Detection

Sysmon Event ID 1 represents Process Creation. Useful fields include Image, CommandLine, ParentImage, User, and ProcessId.

Detection Query

```
index=sysmon EventCode=1
| table _time User Image CommandLine ParentImage ProcessId
```

Processes Worth Investigating

```
powershell.exe
cmd.exe
wscript.exe
cscript.exe
mshta.exe
rundll32.exe
regsvr32.exe
certutil.exe
```

15. Sysmon Network Connection Detection

Sysmon Event ID 3 represents Network Connection. It can help identify suspicious outbound connections, unexpected ports, and possible malware communications.

Detection Query

```
index=sysmon EventCode=3
| table _time Image User SourceIp SourcePort DestinationIp DestinationPort Protocol
```

16. Account Creation Detection

Windows Event ID 4720 indicates that a new user account was created.

Detection Query

```
index=windows EventCode=4720
| table _time SubjectUserName TargetUserName host
```

17. Account Deletion Detection

Windows Event ID 4726 indicates that a user account was deleted.

Detection Query

```
index=windows EventCode=4726
| table _time SubjectUserName TargetUserName host
```

18. Privileged Account Activity

Windows Event ID 4672 indicates special privileges were assigned to a new logon.

Detection Query

```
index=windows EventCode=4672
| table _time SubjectUserName PrivilegeList host
```

19. Process Creation – Windows Event ID 4688

Windows Event ID 4688 records a new process being created and can support process-execution investigations.

Detection Query

```
index=windows EventCode=4688
| table _time New_Process_Name Creator_Process_Name SubjectUserName
```

20. IOC Investigation

Splunk can be used to search for indicators of compromise such as IP addresses, domains, file hashes, and URLs.

IOC Queries

```
index=* "192.168.1.100"

index=* "example.com"

index=* "SHA256_HASH"

index=* "http://example.com"
```

21. Detection Engineering

A SOC detection generally contains: data source, detection logic, threshold, time window, severity, alert condition, investigation steps, and response action.

Example – Brute Force Detection

Data Source: Windows Security Logs
Event: 4625
Condition: 5 or more failures
Time Window: 5 minutes
Severity: Medium/High depending on context

Detection Query

```
index=windows EventCode=4625
| bin _time span=5m
| stats count BY _time src_ip User
| where count >= 5
```

22. Alert Creation

A Splunk alert can be created from a detection search. Example configuration:

Alert Name: Multiple Failed Login Attempts
Trigger: Number of results > 0
Schedule: Every 5 minutes
Severity: Medium
Action: Notify SOC analyst or create an incident ticket.

23. SOC Dashboards

Useful dashboards include Authentication Dashboard, Endpoint Security Dashboard, and Incident Dashboard.

Authentication: successful logins, failed logins, top users, top source IPs, RDP logins.
Endpoint: process creation, PowerShell execution, network connections, suspicious processes.
Incident: open alerts, critical alerts, high-severity incidents, incident status, and response metrics.

24. SOC Investigation Workflow

Alert Generated → Initial Triage → Identify User / Host / IP → Search Related Events → Analyze Timeline → Check Threat Intelligence → Determine True Positive / False Positive → Document Findings → Create / Update Jira Ticket → Escalate or Close

25. Example Investigation – Multiple Failed Logins

Scenario: The SOC receives an alert for multiple failed login attempts.

Step 1: Search failed logins.
Step 2: Identify the suspicious source IP.
Step 3: Check for successful login activity.
Step 4: Review RDP, PowerShell, process creation, account changes, and network connections.
Step 5: Validate relevant IOCs with threat intelligence.
Step 6: Determine severity and escalate if malicious.
Step 7: Document the alert, findings, evidence, and response.

26. Key Skills for a SOC Analyst

Splunk SIEM, SPL, Windows Event Logs, Sysmon, authentication monitoring, brute-force detection, PowerShell monitoring, process analysis, IOC investigation, alert triage, threat hunting, dashboards, reporting, and incident documentation.

27. Conclusion

Splunk is an important SOC platform for collecting and analyzing security logs. A SOC Analyst L1 should be able to search security events, understand Windows Event IDs, write basic SPL queries, filter and aggregate events, investigate suspicious activity, create detections and alerts, build dashboards, perform IOC searches, and document incidents.

Key principle: Do not focus only on writing SPL. Understand what the detected activity means, why it is suspicious, and what the SOC analyst should do next.